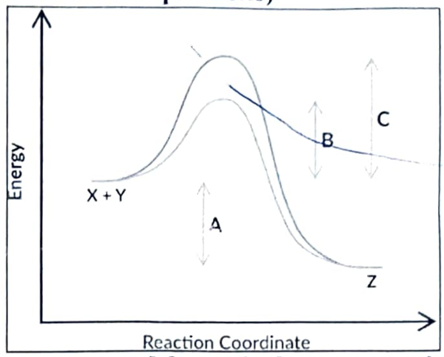
IB

sem I

Minor Examination [Introduction to Bioengineering Theory, BBL1020] [Set A]

Duration: 2 hours **Total marks: 40** **20/09/2024**

Multiple choice questions, each question carries 1 mark (Please answer all the questions)



Q1. In this reaction coordinates graph, which of the following statements about letter *B* is *true*?

A. Letter B represents the energy difference between the reactants and products for the uncatalyzed pathway.
B. Letter B represents the energy difference between the reactants and products for the catalyzed pathway.
C. Letter B represents the activation energy of the uncatalyzed reaction pathway.
D. Letter B represents the activation energy of the catalyzed reaction pathway.

Q2. You have analyzed the genome isolated from a newly discovered virus and found its base composition as Adenine = 32%, Thymine = 19%, Guanine = 32%, and Cytosine = 17%.
What would be a reasonable explanation for this observation?

(A) DNA virus with higher purines, **(B)** DNA virus with higher pyrimidines
(C) RNA virus with higher purines, **(D)** RNA virus with higher pyrimidines

Q3. Miller's experiment in 1952 showed that

(A) Energy is required for chemical reactions, **(B)** Organic compounds can be synthesized from abiotic matter, **(C)** Life on earth originated from extraterrestrial events, **(D)** All of these

Q4. Evolution is based on which of the following concepts?

(A) Organisms share a common origin, **(B)** Over time, organisms have diverged from a common ancestor, **(C)** An animal's body parts can change over its lifetime, and these acquired changes are passed on to the next generation, **(D)** Both 1 and 2, **(E)** All of the above.

Q5 Which of the following is an enzyme that catalyzes the breakdown of hydrogen peroxide?

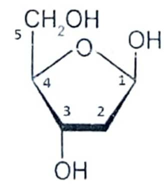
(A) Amylase, **(B)** Catalase, **(C)** Dehydrogenase, **(D)** Lipase

Q6 Which of the following is an example of artificial evolution/selection?

(A) Natural selection of longer neck Giraffe, **(B)** Selective breeding for higher milk producing cow, **(C)** Survival of camel with a hump on back, **(D)** Finches developing different beak sizes

Q7 Which of the following reactions has a negative Gibbs free energy (ΔG)?

(A) Photosynthesis, **(B)** Respiration, **(C)** ATP synthesis **(D)** Protein synthesis



Q8. Which carbons of the ribose in nucleic acids bond with phosphate (PO_4)?

(A) 1 & 3, **(B)** 1 & 5, **(C)** 3 & 5, **(D)** All of these

Q9. The process of evolution is driven by

(A) Genetic variations, **(B)** Environmental stress, **(C)** Reproduction ability
(D) All of these, **(E)** None of these

Q10. What property of water makes an excellent solvent?

(A) Cohesiveness **(B)** High surface tension **(C)** High specific heat. **(D)** Polar nature

Q11. If a double-stranded DNA has 10% adenine, what percentage of cytosine will be present?

(A) 10% (B) 20%, (C) 30%, (D) 40% (E) none of these

Q12. If one tablespoon of glucose produces 4 kilocalories, how much energy will a tablespoon of cooking oil produce?

(A) Less than 4 kilocalories **(B)** 4 kilocalories, **(C)** more than 4 kilocalories **(D)** no calories

Minor Examination [Introduction to Bioengineering Theory, BBL1020] [Set A]

Duration: 2 hours Total marks: 40 20/09/2024

Q13. Glycogen is mostly used as
(A) Everyday energy (B) Stored energy. C) Building material for a cell. (D) Instant energy

Q14. Which type of protein structure is characterized by alpha helices and beta sheets?
(A) Primary structure. (B) Secondary structure. (C) Tertiary structure. (D) Quaternary structure

Q15. What type of function do nucleotides perform in a cell?
(A) Storage of information (B) Storage of energy (C) Cellular signaling. (D) All of them

Q16. Please identify the molecule with catalytic activities among the molecules given below.
(A) Protein. (B) Carbohydrates. (C) DNA. (D) Lipid

Short answer, each question carries 2 marks (Please answer any 6 questions)

Q17. What type of lipids were involved in the origin of life and how they contributed to that process?

Q18. Haber's reaction is the traditional way to produce ammonia from N2 and H2. This reaction requires high temperature and pressure. Please explain if you can produce ammonia without using high temperature and pressure.

Q19. Propanol is toxic to us. Using structure-activity relationship knowledge, how can you make it safer?

Q20. What are the phenotypic variations within a species and why they are important?

Q21. What properties of the Carbon atom make it important for life on planet Earth?

Q22. Let's assume you have two fragments of double-stranded DNA. Fragment X contains 20% Adenine and Fragment Y contains 40% Adenine. Which fragment will require more energy to convert into a single-strand molecule and why?

Q23. How does ATP function as an energy carrier in biological systems, and what role does it play in energy coupling?

Q24. CO_2 does not readily dissolve in water, hence, cannot be transported within the body efficiently. How does the body quickly remove CO_2?

Long Answer, each question carries 4 marks (Answer any 3 questions)

Q25. Please describe with an example how enzyme engineering can create products that fight mitochondrial diseases.

Q26: Please describe the energy coupling and thermodynamics of a cellular system that keeps it running continuously.

Q27. Please describe with two examples, how biocatalysts have benefited drug manufacturing companies.

Q28. Please describe, using an example, how different pharmacophores alter the structure-activity relation (SAR) of a compound.

B24CM1063

Sem I

Minor Examination [Introduction to Bioengineering Theory, BBL1020] [Set B]

Duration: 2 hours **Total marks: 40** **20/09/2024**

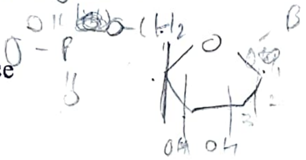
Multiple choice questions, each question carries 1 mark (Please answer all the questions)

Q1. Which of the following is true for catalytic enzymes?
I. They increase the rate of reaction by stabilizing the transition state.
II. They raise activation energy to shift the equilibrium to favor the products.
III. They lower activation energy by altering the products of a reaction.
(A) I only, **(B)** II + III, **(C)** III only, **(D)** I + III, **(E)** II only

Q2. The nucleotide composition of the virus X genome is Adenine 36.2%, Thymine 28.2%, Guanine 19.9%, and Cytosine = 15.7%. As per this information, the virus X genome is made up of:
(A) Single-stranded RNA (ssRNA), **(B)** Double-stranded DNA (dsDNA), **(C)** Single-stranded DNA (ssDNA), **(D)** Double-stranded RNA (dsRNA), **(E)** It cannot be determined

Q3. In the following question, statement of Assertion (A) is followed by statement of Reason (R).
Assertion (a): In a DNA molecule, A-T rich regions undergo melting prior to G-C rich regions.
Reason (r): In between A and T, there are three H-bonds, whereas in between G and C, there are two H-bonds.
Choose the correct answer from the options given below.
(A) Both (a) and (r) are true, and R is the correct explanation of (a). **(B)** Both (a) and (r) are true, but R is NOT the correct explanation of (a). **(C)** (a) is true, but (r) is false. **(D)** (a) is false, but (r) is true. **(E)** Both (a) and (r) are false.

Q4. A new drug is developed that selectively cleaves covalent bonds between two sulfur atoms of non-adjacent amino acids in a polypeptide chain. Which level/(s) of protein structure in the affected molecules would be most directly influenced by the drug?
(A) Primary structure, **(B)** Secondary structure, **(C)** Tertiary structure, **(D)** All of these



Q5. Where is a hydrogen bond present within a nucleotide?
(A) Between phosphate and 3'ribose **(B)** Between phosphate and 5'ribose
(C) Between nitrogenous base and 1'ribose **(D)** Not present

Q6. What bonds provide the DNA its double helical structure?
(A) Phosphodiester bond, **(B)** Glycosidic bond, **(C)** Hydrogen bond, **(D)** All of the above

Q7. Use of catalyst in a reaction does not change its ....
(A) kinetics **(B)** activation energy **(C)** Thermodynamics **(D)** time to complete

Q8. Which macromolecule can be made using the micromolecules given below-
1. Nitrogenous base, 2. Fatty acids, 3. Phosphate, 4. Glycerol, 5. Glucose
(A) Protein, **(B)** Lipid, **(C)** DNA **(D)** RNA, **(E)** Carbohydrate

Q9. What property of water makes it an excellent heat retainer?
(A) Cohesiveness, **(B)** High surface tension, **(C)** High specific heat, **(D)** Polar nature

Q10. Which of the following reactions is exergonic with a $\Delta G < 0$?
(A) Photosynthesis, **(B)** ATP hydrolysis, **(C)** Protein synthesis, **(D)** Glucose phosphorylation

Q11. If a double-stranded DNA has 20% Guanine, what percentage of Adenine will be present?
(A) 10% **(B) 20%,** **(C) 30%,** **(D) 40%** **(E) none of these**

Q12. Evolution is based on which of the following concepts?
(A) Organisms share a common origin, **(B)** Over time, organisms have diverged from a common ancestor, **(C)** An animal's body parts can change over its lifetime, and these acquired changes are passed on to the next generation, **(D)** Both 1 and 2, **(E)** All of the above.

Set B

Minor Examination [Introduction to Bioengineering Theory, BBL1020] [Set B]

Duration: 2 hours **Total marks: 40** **20/09/2024**

Change?

Q13. Gibbs free energy for photosynthesis is
(A) Less than 0. **(B)** 0. **(C)** Greater than 0. **(D)** Difficult to calculate

Q14. Which of the following is NOT a function of proteins?
(A) Catalyzing biochemical reactions, **(B)** Storing genetic information, **(C)** Cellular signaling, **(D)** Transportion

Q15. Lipids are hydrophobic, but phospholipids have hydrophilic properties as well. Because they have
(A) Longer chain of fatty acids, **(B)** Glycerol, **(C)** Phosphate, **(D)** Sugar, **(E)** Amino acid

Q16. Adenosine triphosphate (ATP) is our-
(A) Genetic information molecule, **(B)** Energy molecule, **(C)** signaling molecule, **(D)** all of them

Short answer, each question carries 2 marks (Please answer any 6 questions)

Q17. It is believed that the life is originated in the water. Please explain the evidences explaining this theory.

Q18. Define the type of catalysts with examples.

Q19. Phenol is toxic to us. Using structure-activity relationship knowledge, how can you make it safer?

Q20. How does evolution contribute to the theory of "survival of the fittest"?

Q21. Why hydrogen bonds of water molecules are important for sustaining life on planet Earth?

Q22. Let's assume you have two fragments of double-strand DNA. Fragment X contains 30% Guanine and Fragment Y contains 20% Guanine. Which fragment will require more energy to convert into a single-strand molecule and why?

Q23. What are ribozymes, and how do they differ from traditional enzymes?

Q24. Our mouth cavity is constantly exposed to the germs. How does the saliva circulating the mouth help us prevent germ infections?

Long Answer, each question carries 4 marks (Answer any 3 questions)

Q25. Please describe with an example how enzyme engineering can create products that fight mitochondrial diseases.

Q26. Please describe the energy coupling and thermodynamics of a cellular system that keeps it running continuously.

Q27. Please describe with two examples, how biocatalysts have benefited drug manufacturing companies.

Q28. Please describe, using an example, how different pharmacophores alter the structure-activity relation (SAR) of a compound.

SEM I

Major Examination [Introduction to Bioengineering Theory, BBL1020]

Set B Duration: 3 hours Total marks: 80 20/11/2024

Q1-26, each question carries 1 mark (Please answer all the questions)

Q1. Which of the following is an example of paracrine signaling?
A. Epinephrine signaling in muscle cells B. IL2 signaling in T cells C. Neurotransmitter signaling in Neurons D. All of these

Q2. Which of the following molecules activate G-protein in the process of cell signaling?
A. ATP B.GTP C. TTP D. CTP

Q3. Which one of these is a common stress factor in the bioproduction industry?
A. Stirring B. High passage number C. Local nutrient deprivation D. All of these

Q4. Which one of the following microorganisms **cannot** be used to produce proteins in Biotech industries?
A. Bacteria B. Yeast C. Algae D. Virus

Q5. What type of bond stabilizes the secondary structure of proteins?
A. Peptide bond B. Hydrogen bonds C. Covalent bonds D. Disulfide bridges

Q6. In an operon, what is the role of the operator region?
A. Binds RNA polymerase B. Produces enzymes C. Controls transcription D. Synthesizes RNA

Q7. Which of the following can be used for carbon fixation?
A. Photobioreactors B. Plantation C. Thermal reactors D. Both (A)&(B) E. Both (B)&(C)

Q8. What is the drawback of second-generation biofuel?
A. Directly interferes with our food supply chain B. Indirectly interferes with our food supply chain C. Generates less efficient biofuel D. All of these

Q9. What is the function of the Arabinose operon in a bacterial cell?
A. Avoid unnecessary energy loss B. Digesting food C. Controlling gene expression D. All of these

Q10. Restriction enzymes of a bacteria are part of its
A. Digestive system B. Stress response system C. Defense system E. All of these

Q11. Cloning site in a plasmid vector allows you to-
A. Insert a desired gene B. Cut the circular vector C. Generate sticky ends D. All of these

Q12. Disk diffusion test helps us detect-
A. The diffusion rate B. Antibacterial activity C. Strength of antibiotics D. Both (A) and (B) E. Both (B) & (C)

Q13. Which of the following proteins activates the anaphase-promoting complex (APC/C)?
A. Separase B. Cdc25 C. Cdc20 D. Mad2

Q14. Antibiotics are an example of which of the following types of biological interactions-
A. Competition B. Antagonism C. Mutualism E. Parasitism

Set B

Q15. Which class of organisms will thrive in a soda lake?
A. Neutralophiles B. Acidophiles C. Alkaliphiles D. None of the above

Q16. What type of cell wall does a gram-positive bacterium have?
A. Thick peptidoglycan B. Thin peptidoglycan C. Two phospholipid bilayers D. LPS layer

Q17. *E. coli* is a facultative anaerobe that uses oxygen and lower the O2 concentration in gut which creates suitable environment for obligate anaerobes such as *Bacteroides*. This is an example of-
A. Parasitism B. Communalism C. Antagonism D. Mutualism

Q18. Rb protein helps the cells to stay in which of the stage of the cell cycle
A. G1 B. S C. G2 D. M E. G0

Q19. In tissue engineering, what is the purpose of incorporating growth factors such as VEGF (vascular endothelial growth factor) into scaffolds?
A. To stimulate cell adhesion and migration
B. To promote the formation of blood vessels
C. To prevent immune rejection
D. To increase the mechanical strength of the scaffold

Q20. Which of the following is an example of a positive feedback mechanism?
A. Glucose levels cause a release of insulin, which decreases glucose levels
B. The hair on an elk stands up when the skin gets cold, trapping heat and warming the skin.
C. Glucagon is released by the pancreas when glucose levels are low, causing the liver to release glucose and raise blood glucose levels.
D. A man starts to choke on a hotdog. This further stimulates his gag reflex and eventually leads to him vomiting, which dislodges the hotdog.

Q21. Horizontal gene transfer is related to
A. Stress response B. Drug resistance C. Recombinant DNA technology D. Cell signaling

Q22. A bacterium expressing beta-lactamase enzyme will not be killed by which of the following antibiotics-
A. Penicillin B. Chloramphenicol C. Tetracycline D. Fluoroquinolone

Q23 to Q26, True/False type questions

Q23. A particular gene transcribes the mRNA in the Nucleus, which gets transported to the cytoplasm for translation. Is the sequence of mRNA of a particular gene the same in the Nucleus and cytoplasm?

Q24. The mechanical properties of a scaffold do not depend on the mechanical properties of the material used but on the structure of the scaffold.

Q25. Bioprinting is a type of additive manufacturing.

Q26. Negative feedback mechanisms help us maintain the physiological range of body temperature.

11/20

Major Examination [Introduction to Bioengineering Theory, BBL1020]

Set B | Duration: 3 hours | Total marks: 80 | 20/11/2024

Q27 to 39, Short answer questions. Each question carries 3 marks. Attempt only 12 questions of your choice.

Q27. Hydrocarbons of crude oil and carbohydrates of biomass are both long carbon chain compounds. However, both of them possess very different properties. Please make a table of chemical and structural differences between these two substances.

Q28. Which class of bacteria among gram-positive and gram-negative is often pathogenic and why?

3

Q29. Describe the role of sticky ends in recombinant DNA technology.

2

Q30. What is a plasmid? What is its industrial use?

1

Q31. Consider a genetic circuit designed to express a reporter gene (GFP) only in the presence of a specific environmental signal (As). How would you optimize such a circuit to prevent false positives due to leaky expression?

1

Q32. How does a cell ensure that it starts the cell division only after it gains an optimum size?

2

Q33. How do mutations affect genetic information and lead to variation?

1

Q34. When we consume tetracycline antibiotic drugs, it kills the pathogenic bacteria but not our cells. Why?

2

Q35. Match the proteins with the type of stress response they are associated with.

3

	Name of Proteins		Type of Stress
1	Hkr1, Msb2, HOG1	A	pH stress
2	P53, ATM, ATR	B	Heat stress
3	HSP, Chaperons	C	Osmotic stress
4	Proton pumps	D	UV radiation stress

Q36. Suppose you are producing a secretory protein X in the experimental human cell. This protein removes blood clots from the blood circulation. However, the protein yield is significantly lower than the expected amount. Further analysis detected Endoplasmic Reticulum (ER) stress in the cells. Please explain the reason for this observation and a possible strategy to overcome this problem.

1

Q37. How we can convert the carbohydrates into hydrocarbons, that can be used as fuel? Please explain the challenges associated with these processes.

3

Q38. Does the gene expression occur at a constant rate? Please explain using an example of the prokaryotic cell.

2

Q39. What role does the protein p21 play in the regulation of the cell cycle?

3

Q40 to 43, Long answer questions. Each question carries 6 marks. Attempt only 3 questions of your choice.

Major Examination [Introduction to Bioengineering Theory, BBL1020]

Set B Duration: 3 hours Total marks: 80 20/11/20

Q40. Given a gene that codes for a therapeutic protein P, outline the steps you would take to create a transgenic bacterium capable of producing protein P.

Q41. Discuss the mechanisms of cell cycle arrest at the G1, S, and G2 checkpoints in response to DNA damage. How do the tumor suppressor proteins p53 and Rb regulate these checkpoints, and what are the consequences of their malfunction?

Q42. Pathogenic bacteria evolve to develop antibiotic resistance. Please describe various mechanisms adopted by the bacteria to resist different classes of antibiotics.

Q43. The cells are equipped to interpret the signals received from the surroundings and act in response to those signals. Please explain this process with an example related to the cell cycle progression/regulation.

0/11/20

sem II

BBL1020: Introduction to Bioengineering Theory

Minor Examination

Date: **23/02/2025**; Total **Marks=20**; Time: **2:00-4:00** pm

INSTRUCTIONS:

- **Attempt all questions**
- **Write your name & roll no. on the answer sheet. Tie and secure any extra answer sheets properly.**
- Use of **scientific calculator is allowed. However, calculator sharing is not allowed**

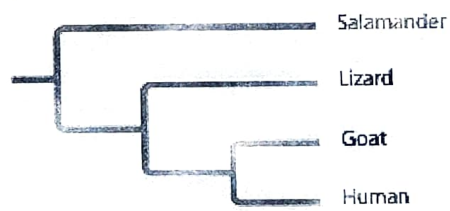
Choose only ONE correct answer from the provided choices. (1 mark each)

Question 1: The pH of "Kalyani Lake" in Jodhpur is 8. What is the hydroxide ion concentration of the lake.

(A) 10^{-8} M (B) 10^{-7} M (C) 10^{-6} M (D) 10^{8} M

pH = 8 $[H^+] = [10^{-8}]$

Question 2 Based on the tree below, which statement is not correct?



(A) Goats and humans form a sister group.
(B) Salamanders are a sister group to the group containing lizards, goats, and humans.
(C) Salamanders are as closely related to goats as to humans.
(D) Lizards are more closely related to salamanders than to humans.

Question 3: The structural level of a protein least affected by a disruption in hydrogen bonding is the

(A) primary level. (B) secondary level. (C) tertiary level. (D) quaternary level.

Question 4: Which of the following factors would tend to increase membrane fluidity?

(A) a greater proportion of unsaturated phospholipids

(B) a greater proportion of saturated phospholipids

(C) a lower temperature

(D) a relatively high protein content in the membrane

Question 5: Which of the following statements correctly describes any chemical reaction that has reached equilibrium?

(A) The concentrations of products and reactants are equal.

(B) The reaction is now irreversible.

(C) Both forward and reverse reactions have halted.

(D) The rates of the forward and reverse reactions are equal.

$5 = 4.76 + \ln [\ ]$

$0.24 = \ln_e [\ ]$ $e^{0.24}$

Question 6: Calculate the ratio [CH_3COOH]/[$NaCH_3COO$] that gives a solution with pH = 5.00? (pKa of acetic acid is 4.76)

(A) 0.28 (B) 0.36 (C) 0.44 (D) 0.56 (E) 0.63

Short Answer type (2 Marks each)

Question 7: Imagine a protein that functions in the ER but requires modification in the Golgi apparatus before it can achieve that function. Describe the protein's path through the cell, starting with the mRNA molecule that specifies the protein.

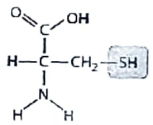
Question 8: Males afflicted with Kartagener's syndrome are sterile because of immotile sperm, and they tend to suffer from lung infections. Suggest which cell organelle is most likely to be underlying cause of this defect and why.

Question 9: Draw and name the chemical structure for two monosaccharide that has three carbons.

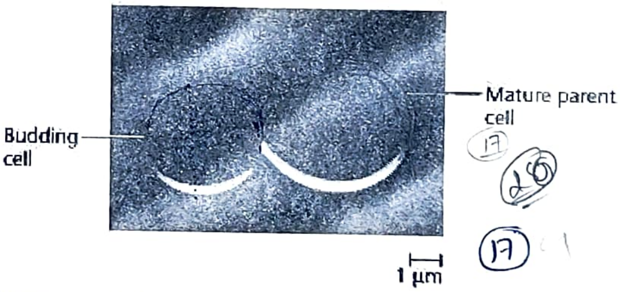
Question 10: Where would you expect a polypeptide region rich in the amino acids, valine, leucine, and isoleucine to be located in a folded polypeptide? explain.

Question 11: If you place a teaspoon of sugar in the bot tom of a glass of water, it will dissolve completely over time. Left longer, eventually the water will disappear, and the sugar crystals will reappear. explain these observations in terms of entropy.

Question 12: Here is the chemical structure of a sulphur containing amino group, Cysteine. Suppose you remove the -NH2 group of cysteine amino acid and replace it with a carboxyl group. How would this change the chemical properties of the molecule? State at least two changes in properties.



Question 13: The unicellular yeast Saccharomyces cerevisiae divides by budding off a small new cell that then grows to full size. During its growth, the new cell synthesizes new cytoplasm, which increases its volume, and new plasma membrane, which increases its surface area. The shape of a yeast cell can be approximated by a sphere.



(a) Calculate the volume (state units) of each cell using the formula for the volume of a sphere:

Hint: The volume of sphere is $\frac{4}{3}\pi r^3$ (π= 3.14, r stands for radius, which is half the diameter).

(b) What volume (state units) of new cytoplasm will the new cell have to synthesize as it matures?

Hint: To determine this, calculate the difference between the volume of the full-sized cell and the volume of the new cell.

$\frac{20}{4}$ 5

Sem II

BBL1020: Introduction to Bioengineering Theory

Major Examination

Date: 27/04/2025; Total Marks=40; Time: 2:00-5:00 pm

INSTRUCTIONS:

- **Attempt all questions**
- **Write your name & roll no. on the answer sheet. Tie and secure any extra answer sheets properly.**
- **Use of scientific calculator is allowed. However, calculator sharing is not allowed**

Choose only ONE correct answer from the provided choices. (1 mark each)

Question 1: Transcription occurs along a template read in____ direction, forming an mRNA in the _____ direction.

Am (270

A. 5' to 3'; 5' to 3'
B. 5' to 3'; 3' to 5'
C. 3' to 5'; 5' to 3'
D. 3' to 5'; 3' to 5'

Question 2: Transcription and translation of a gene composed of 30 nucleotides would form a protein containing no more than ___ amino acids.

A. 10 B. 15 C. 60 D. 90

Question 3: During translation, the _____ site within the ribosome holds the growing amino acid chain while the _____ site holds the next amino acid to be added to the chain.

A. A, P
B. P, A
C. A, B
D. B, A

Question 4: In humans, brown eyes (B) are dominant over blue (b). A brown-eyed man marries a blue-eyed woman and they have three children, two of whom are brown-eyed and one of whom is blue-eyed. What must the genotype of the man be?

A. BB B. Bb

Question 5: Sharon has 100 rabbits. Seventy-five of her rabbits are black and twenty-five of the rabbits are white. Based on this information, what can she conclude about the rabbits?

A. The parents of the rabbits were homozygous dominant
B. The genotypes of the rabbits were Rr x RR
C. One parent was homozygous dominant and one parent was recessive
D. The genotypes of the rabbits were Rr x Rr

100

Question 6: An enzyme that joins the ends of two strands of nucleic acid is:

A. Polymerase
B. ligase
C. synthetase
D. Helicase

Question 7: The enzyme minus its coenzyme known as:

A. Apoenzyme

B. Metalloenzyme

C. Isoenzyme

D. All of these

Question 8: Transcription in prokaryotes starts on which site?

A. Regulator B. RNA polymerase C. Initiator D. Promoter

Question 9: A mutation that leads to a nucleotide sequence to change the corresponding amino acid in the protein chain is called?

A. Silent Mutation B. Missense Mutation C. Nonsense Mutation D. Frameshift Mutation

Question 10: A solution of starch at room temperature does not readily decompose to form a solution of simple sugars because:

A. the starch solution has less free energy than the sugar solution.
B. the hydrolysis of starch to sugar is endergonic.
C. the activation energy barrier for this reaction cannot be surmounted.
D. starch cannot be hydrolyzed in the presence of so much water.
E. starch hydrolysis is nonspontaneous.

Question 11: Write TRUE or FLASE for the given statements: (0.5x6=3 marks)

I. The lagging strand is the new strand that grows discontinuously away from the replication fork.
II. A codon may code for the same amino acid as another codon does.
III. The organism ultimately must obtain all the necessary energy for life from its environment.
IV. Centromere is located exactly at the center of the chromosomes.
V. The bacterial gene is interspersed by introns.
VI. Both DNA polymerase and RNA polymerase require a primer

Question 12: Write a step from Glycolysis which generates ATP through substrate level phosphorylation. (1 marks)

1 2

Question 13: If you oxidize fats and carbohydrates.

a) Which one of them will release more energy?
b) Explain your choice in (a)?

(2marks)

Question 14: Pyruvate generated by Glycolysis is converted to acetyl-coenzyme A, which is metabolized by the citric acid cycle to generate energy-rich molecules and gases. Write the number of such molecules produced from one molecule of acetyl-coenzyme A

a) GTP b) NADH c) $FADH_2$ 2 d) CO_2 (2 marks)

Question 15: For a coupled reaction given below, the equilibrium constant (Keq) for equation [1] and [2] are 270 and 890, respectively:

Glucose 6-phosphate $+H_2O \rightarrow$ Glucose $+Pi$ [1] Kg = 270

ATP + Glucose $\rightarrow$ ADP + Glucose 6-phosphate [2] Kg = 890

What is the standard free energy change of ATP at 25 °C? Give your answer in KJ/mol (2 marks)

Question 16: A eukaryotic structural gene has two introns and three exons:

5'-exon1-intron1-exon2-intron2-exon3-3'

3

5

$K = K_1 \times K_2$

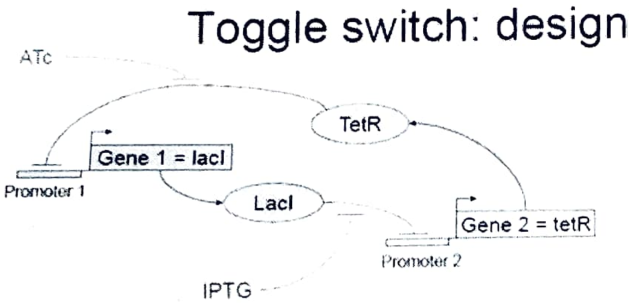
The GU at the 5' end of intron2 has been mutated so it is no longer recognized. What would the mature mRNA look like in the wild type and in the mutant? (2 marks)

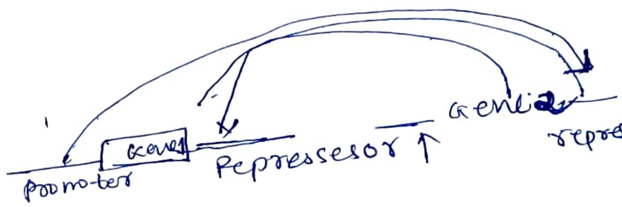
Question 17: A heterozygous round seed plant is crossed with a heterozygous round seed plant. What percentage of the offspring will be homozygous? Explain (2 marks)

Question 18: You have added an inhibitor that blocks electron flow through complex 1 of the electron transport chain in mitochondria. (2 Marks)

a) The ATP synthesis will (choose the appropriate option below):
   i. Remain same ii. Increase iii. Decrease
b) Explain your choice in (a).

Question 19: The diagram below shows how the gene circuit shown in the figure simulates a toggle switch (2 Marks)



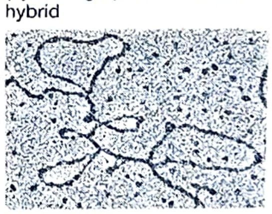


If the two options for the toggle switch are either Gene 1 or 2 being turned on.

a) Briefly explain which gene ATc turns on, and why?
b) Briefly explain which gene IPTG turns on, and why?

Question 20: A geneticist isolates a gene from eukaryotic cells and then isolates the mature mRNA produced by this gene. After making the DNA single stranded, she mixes the single-stranded DNA and RNA. Some of the single-stranded DNA hybridizes (pairs) with the complementary mRNA as seem by electron microscopy (figure). (2 marks)

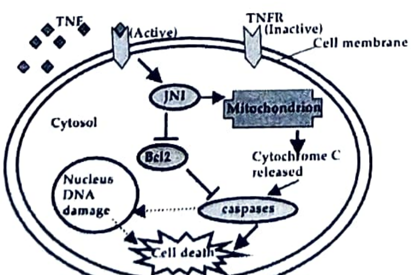
(a) Micrograph of DNA-RNA hybrid



a) What are the loops protruding from the ssDNA-mRNA hybrid? ssDNA or mRNA?

b) Why are there protruding loops when ssDNA-mRNA are hybridized?

Question 21: Consider the following schematic of a signal transduction pathway that results in apoptosis (programmed cell death).



- Tumor necrosis factor (TNF) binds to and activates its specific receptor (TNFR)
- The activated TNFR activates JNI.
- The activated JNI activates caspases by releasing cytochorome C from mitochondria and by inhibiting Bcl2. Bcl2, when active, promotes cell survival by inhibiting caspases.
- The active caspases result in DNA damage and cell death.
- In the schematic an arrow represents activation and a T represents inhibit

Based on the signaling pathway shown in the schematics, what would be the effect of TNF on the cell that shows a loss-of function mutation of the following proteins? (3 Marks)

Cell showing loss of function mutation of	Increased/decreased cell death compared to TNF-treated control wild-type cell?
TNFR	
Bcl2	
Caspases	

Question 22: Some cells in the heart will release a chemical when they are over-stretched. This chemical is transported in the plasma and acts on cells in the kidney to deactivate or decrease sodium transepithelial transport. Answer the following questions about this regulatory pathway. (3 Marks)

I. The signalling chemical is an example of a(n)
   a) Autocrine b) paracrine c) hormone d) neurotransmitter

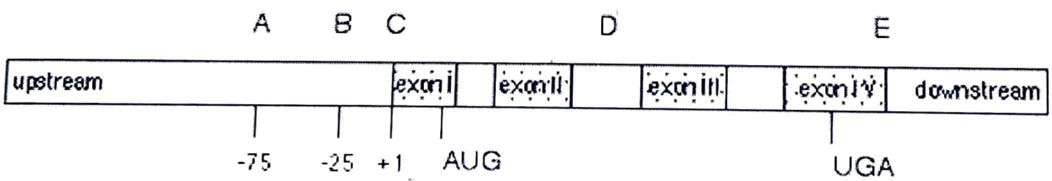
II. The signalling chemical is a protein. You can expect, therefore, that it acts through membrane-spanning or intracellular receptors.

1. A drug has been made that bind to and activates the receptor. This drug is a(n) (choose your choice)

   a) agonist b) antagonist c) isotope d) isoform

2. State a possible mechanism through which the signalling molecule can decrease sodium transepithelial transport.

Question 23: This diagram represents regions around a gene encoding a mRNA. Exon I of the mRNA contains the translation initiation AUG codon. Exon IV contains the UGA termination codon. The letters A through E above the diagram refer to areas of the gene and its transcript located directly below them. Answer the question below by indicating the choice of region using provided letter code (4 marks)



a) What region of the primary transcript is removed by a spliceosome?
b) What region of the DNA contains the promoter?
c) Where is the poly (A) tail added to the transcript?
d) Where is the 5' cap added to the transcript?